1.	Name of Course: **Data Mining Method in the Network Era**
2.	Course Code: **G0303**
3.	Name(s) of Academic Staff: **Longbiao CHEN**
4.	Rationale for the Inclusion of the Course in the Programme **This course teaches state-of-the-art data mining method in the network era. It aims at providing students with the basic knowledge of data mining, data mining techniques and some useful tools. Student will benefit from this course both in their future study and work.**
5.	Semester and Year Offered: **201702**

6.

Total Student Learning Time (SLT)	Face-to-Face				Total Guided and Independent Learning
L = Lecture T = Tutorial P = Practical O= Others	L **30**	T	P **15**	O	**Independent = 75** **Total = 120**

Details of Total Guided and Independent Learning:

	Learning Activities	SLT (in hours)
1	**Lectures**	**60**
a	Attending Lectures	30
b	Pre and Post Preparation	30
2	**Tutorial**	
a	Attending Tutorial	
b	Preparation for Tutorial	
3	**Laboratory / Field work**	**30**
a	Practical	15
b	Prepreparation and Report Writing	15
4	**Assessments**	**30**
a	3 Assignments (1 hour + 4 hours preparation)	15
b	1 Final Examination (2 hours + 13 hours preparation)	15
	Total	**120**

7.	Credit Value: **3**
8.	Prerequisite (if any): **None**
9.	Objectives: **1. Providing students with the basic knowledge of data mining techniques.** **2. Teaching the student to master some useful tools in real applications.**
10.	Learning Outcomes: **1. Demonstrate knowledge on basic techniques in classification, clustering, association rules and anomaly detection.** **2. Display ability to apply knowledge in benchmark and real-world datasets using octave/matlab.**

11. Transferable Skills:

Skills	How they are	
	Developed	Assessed
knowledge	**Attending lectures and tutorials, independent learning**	**assignment, examination**
Techniques	**Attending lectures and tutorials , completing assignments and projects given, independent learning**	**assignment, examination**
Critical thinking	**Active participation in independent learning and tutorial sessions**	**Lecturer observations, assignment and examination**

12. Teaching-Learning and Assessment Strategy:

- **Lectures**
- **Assignments**
- **Final examination**

13. Synopsis

Nowadays, data mining has a wide application in many areas. This course mainly introduce the basic concepts and technologies in data mining well as principles and applications of machine learning, pattern recognition and artificial intelligence on data mining. Our goal is to have the undergraduate students to grasp the ideas, technologies and its applications, both in theory and practice.

14. Mode of Delivery: **Lecture, Tutorial**

15. Assessment Methods and Types

- **Assignments (50%)**
- **Final examination (50%)**

16. Mapping of the Course to the Programme Aims

17. Mapping of the Course to the Programme Learning Outcomes

	PLO1	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**
CLO1						√		
CLO2						√		

18. Content Outline of the Course and the SLT Per Topic

Content outline of the course	SLT					
	L	T	P	O	Indep.	Total
1. Introduction and Data Exploration • **What Is Data Mining** • **Motivating Challenges** • **The Origins of Data Mining** • **Data Mining Tasks** • **Types of Data** • **Data Quality** • **Data Pre-processing** • **Measures of Similarity and Dissimilarity** • **Exploring Datasets**	**6**		**3**		**9**	**18**
2. Classification Techniques • **What is a Classifier** • **General Approach** • **Decision Tree Induction** • **Model Overfitting** • **Evaluating Classifiers** • **Rule-Based Classifiers** • **Nearest Neighbour Classifiers** • **Bayesian Classifiers** • **Artificial Neural Network** • **Linear Classifiers**	**6**	**3**			**9**	**18**
3. Association Analysis • **Frequent Item set and Rule Generation** • **Compact Representation of Frequent Item sets** • **Alternative Methods** • **Evaluation of Association Patterns**	**6**	**3**			**9**	**18**
4. Cluster Techniques • **What is Clustering** • **K Means** • **Hierarchical Clustering** • **Other Clustering Methods** • **Cluster Evaluation and Other Issues**	**6**	**3**			**9**	**18**

		5. Anomaly Detection • **Statistical Approaches** • **Proximity-Based Outlier Detection** • **Density-Based Outlier Detection** • **Clustering-Based Techniques**	**6**	**3**			**9**	**18**
		Assessments					**30**	**30**
		Total	**30**	**15**			**75**	**120**
19.	Main References Supporting the Course • **Pang-ning Tang, Michael Steinbach, Vipin Kumar, Introduction to Data Mining. Addison Wesley, 2006.** Additional references supporting the course • **Jiawei Han, Micheline Kamber, Jian Pei. Data Mining: Concepts and Techniques, 3rd Edition, Morgan Kaufmann, 2011.** • **Andrew Ng. Machine Learning. Coursera.org.** **https://www.coursera.org/learn/machine-learning** • **Jiawei Han. Pattern Discovery in Data Mining. Coursera.org** **https://www.coursera.org/learn/data-patterns**							
20.	Other Additional Information							